

When Mass Security Screening Backfires:
Causal Modeling for Complex Equilibrium Systems

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Abstract: Mass screening programs for low-prevalence problems share a common signal-detection structure that the National Academy of Sciences (2003) analyzed for polygraph screening through a single causal pathway: classification accuracy under Bayes' Rule. We extend this to a four-pathway structural causal framework incorporating classification, strategic adaptation, information effects, and resource reallocation, coupled to feedback dynamics requiring σ -separation reasoning (Forré & Mooij 2018) rather than d-separation. The framework formalizes mass screening as performative prediction (Perdomo et al. 2020), addresses within-pathway aggregation via Rubenstein et al.'s (2017) causal consistency criteria, and applies Bayesian uncertainty propagation (Gelman & Carpenter 2020). We analyze three security cases: the EU's proposed Chat Control 2.0 mandatory AI scanning regulation, U.S. police polygraph programs (with novel difference-in-differences reanalysis of nationally representative LEMAS data using negative binomial regression for the zero-inflated count outcome), and the iBorderCtrl facial micro-expression pilot. For Chat Control, all four pathways converge on net harm to child safety: classification backfires under rarity (99.4% false positives under generous assumptions); dedicated-offender evasion exceeds maximum casual-offender deterrence by a factor of 46.6, foreclosing the compensating-subpopulation defense; the information pathway is architecturally absent; and false-positive triage overwhelms investigative capacity by orders of magnitude. The convergence is robust to normative weighting across 27 sensitivity-analysis cells. Police polygraph programs yield empirically ambiguous predictions; iBorderCtrl shows likely net harm, consistent with its abandonment. The recurring failure of mass screening programs across security, medicine, and other domains motivates structural legislative response rather than case-by-case advocacy.

Significance statement: Mass screening programs — from polygraph testing to AI surveillance scanning — share a common signal-detection structure that fails catastrophically under rarity, persistent uncertainty, and secondary screening harms. Existing policy analysis typically focuses on classification accuracy alone, missing how strategic adaptation, information effects, and resource reallocation can either compensate for or amplify classification failure. This paper extends the National Academy of Sciences' 2003 polygraph analysis to a four-pathway causal framework formalizing these dynamics, applied to three high-stakes security cases. For the EU's proposed Chat Control 2.0 regulation, all four pathways converge on net harm to child safety. The framework structures policy debate by identifying which screening programs can plausibly succeed and which cannot.

Keywords: causal inference; mass screening; signal detection; performative prediction; AI surveillance policy; equilibrium modeling

1. Motivation

The National Academy of Sciences' 2003 polygraph report co-chaired by public statistician Stephen Fienberg modeled estimated hypothetical outcomes in terms of true and false positives and negatives along the proposed program's classification causal pathway. This single-pathway analysis applied Bayes' Rule to illustrate why polygraphing all National Lab scientists looking for spies would backfire whether the tests were set to minimize false positives or false negatives due to the accuracy-error trade-off, forcing "an unacceptable choice... between too many loyal employees falsely judged deceptive and too many major security threats left undetected" (NAS 2003, p. 6). This Bayesian backfire and the overarching possibility of net harm from mass screenings of this structure may be counter-intuitive due to the base rate bias (Piattelli-Palmarini, 1994), which often leads people to focus on high advertised accuracy numbers to the exclusion of recognizing the sheer number of false positives and complexity of other effects these programs imply.

Since Fienberg's analysis, causal modeling has become increasingly standard in science (Pearl, 2009; Greenland, Pearl, & Robins, 1999; Glymour & Greenland, 2008). Such modeling reveals three additional pathways through which programs that share this common mathematical structure (signal detection problems; Green & Swets, 1966) may affect outcomes:

1. Classification — test produces true and false positives and negatives (the standard focus)
2. Strategy — behavioral changes in response to the test (deterrence, evasion, gaming, and selection effects)
3. Information — knowledge generated or spread through the screening process itself (elicitation, bogus pipeline, suggestion, and iatrogenesis)
4. Resource reallocation — both quantitative (zero-sum finite capacity) and qualitative (framing effects)

Thus, we cannot estimate such programs' net effects from classification pathway effect estimation alone. Even if the test backfires via this pathway due to the accuracy-error tradeoff under common conditions of (1) rarity of the problem or condition of interest, (2) persistent inferential uncertainty about classification, and (3) secondary screening harms, the strategy and information pathways may still dominate with benefits. Or, along with resource reallocation effects, the false positive backfire along the classification and other pathways may amplify net harm.

Three case studies demonstrate this, ordered by depth of treatment. Chat Control (the EU Regulation on preventing and combating child sexual abuse) receives the most detailed analysis here. The EU Commission proposed mandatory AI scanning of all digital telecommunications in May 2022 (COM(2022) 209 final). The hope was to catch pedophiles transmitting child sexual abuse material (CSAM). After three years of contested negotiations, the Council reached a position in November 2025 (Council of the European Union 2025) that retreated from mandatory scanning of end-to-end encrypted communications (e.g., WhatsApp and Signal). Breyer (2025) documents the legislative timeline.

Current voluntary scanning, which the Commission has proposed extending beyond April 2026, operates on services which are encrypted in transit but unencrypted to the service provider (e.g., Gmail) using hash-matching of known CSAM. Hash-matching of known CSAM operates under fundamentally different statistical logic: it matches against confirmed material rather than classifying unknown content, largely dissolving the base rate problem that drives the mandatory proposal's false positive burden. Critics have raised two strategic behavioral concerns about

perceptual hashing (edge-detection-based content fingerprinting; Marr & Hildreth, 1979): evasion (small image modifications producing large hash value changes, increasing false negatives) and sabotage (adversarial flooding to increase false positives) (Troncoso et al., 2024). These are real concerns, but they do not resurrect the base rate problem. The “security by obscurity” objection — that any system depending on hash secrecy is cryptographically unsound — imports a principle from classical cryptography that pertains to algorithm secrecy, where it is rightly criticized, and misapplies it to hash list secrecy, which serves a different function: protecting against evasion and sabotage rather than providing cryptographic security. Finally, hash matches require physical corroboration — when investigators collect devices, the evidence either holds or it does not — limiting the scope for authoritarian abuse through hash manipulation compared to purely algorithmic classification.

This framework applies specifically to programs that must infer unknown status — e.g., novel abusive content, deceptive intent, or threat classification — from visual, physiological, or behavioral input with probabilistic rather than one-to-one signal versus noise correspondences. Detection of known, pre-specified signals operates under fundamentally different statistical logic, as the hash-matching case illustrates. That distinction defines the scope of what follows: all three case studies involve the former.

The analysis here concerns the previously proposed mandatory AI screening of all digital telecommunications. This analysis matters because member states retain discretion over implementation. Trilogue negotiations between the Council, Parliament, and Commission continue as of this writing, and national-level implementation decisions are expected to be ongoing after their resolution. The current sociopolitical stakes of estimating the net effects of this program are thus the highest of the considered cases. This analysis suggests harm may dominate along every causal pathway, an important finding for child safety.

Police polygraph programs embody a different causal structure: strategy and information effects may dominate due to deterrence and bogus pipeline phenomena, producing ambiguous predictions that a novel empirical analysis attempts to tentatively resolve. This analysis may generalize to mass security screenings with strategic selection effects and interrogation components.

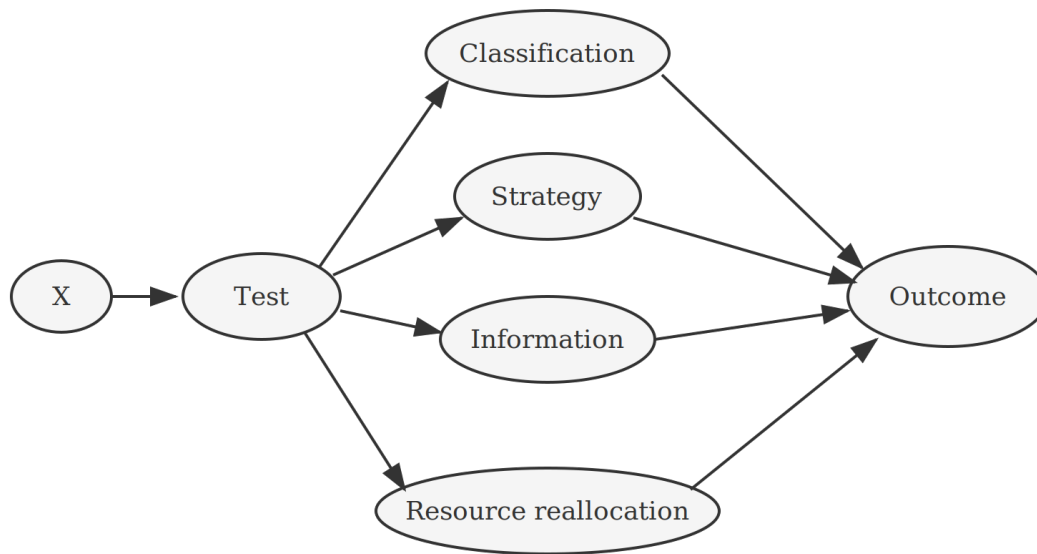
iBorderCtrl receives the most abbreviated treatment; its automated kiosk format has the form of interrogation-based coupling without the substance — lacking adaptive human elicitation — and evidence-based critique prevented deployment beyond the pilot phase. This gradient — from strong possible strategic and information effects in polygraph programs, to attenuated coupling in iBorderCtrl, to none in Chat Control — logically predicts which programs may compensate for classification pathway backfire through other pathways, and which cannot.

Importantly, the framework applies specifically to *mass* screening regimes, where tests are applied population-wide to detect rare conditions. Targeted *investigative* use of similar tools operates under an entirely different statistical structure: base rates are substantially higher and screening occurs after suspicion has already narrowed the population. The causal effects of mass versus targeted screenings may be expected to differ substantially. This application considers only causality in mass screening programs.

Security programs often present special validation challenges that make inferential uncertainty particularly persistent and secondary screening harms particularly grave. The condition being

detected — deceptive intent, future criminal behavior, threat status — is unobservable not just in practice but in principle: behavior is endogenous to the screening itself, so there is no ground truth against which to validate classifications. This means errors may compound silently, and harms to falsely flagged individuals accumulate without any corrective feedback mechanism. Future work will extend this framework to other domains including medicine, where ground truth is often eventually observable and validation problem solutions are more tractable.

2. Multiple Causal Pathways



DAG: Multiple Causal Pathways in Mass Screening

Figure 1. *Mass Screening Directed Acyclic Graph*. The DAG shows multiple causal pathways beyond the typical classification focus are relevant to net program effects.

The four pathways from Test to Outcome:

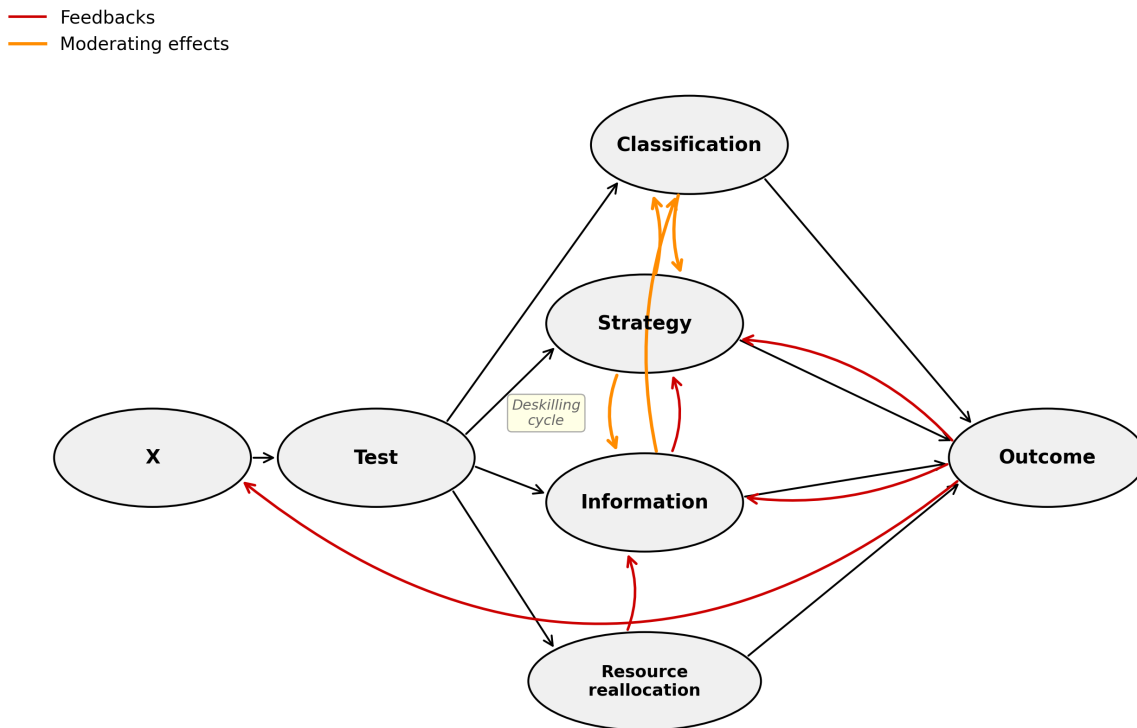
CLASSIFICATION: True Positives, False Positives, True Negatives, False Negatives

STRATEGIC BEHAVIOR: Deterrence, Evasion/Sabotage, Perverse Incentives, Selection

INFORMATION EFFECTS: Elicitation, Suggestion, Iatrogenesis

RESOURCE ALLOCATION: Quantitative (zero-sum), Qualitative (framing)

3. Feedback Loops in Equilibrium



DCG: Multiple Causal Pathways with Feedbacks in Mass Screening

Figure 2. *Mass Screening Directed Causal Graph.* The DCG extends the acyclic model to a directed cyclic graph, incorporating the feedback dynamics (red arrows) — including the deskilling cycle between strategy and information pathways — that motivate the equilibrium modeling approach. It also incorporates moderating effects (orange arrows). For representation and reasoning in cyclic causal models see Ahsan, Arbour & Zheleva (2022). Because the DCG contains feedback loops, conditional independence reasoning requires σ -separation rather than d-separation (Forré & Mooij, 2018) — the cyclic analogue that accounts for equilibrium dependencies.

3.1 Feedback loops

Feedback loops (Outcome → earlier nodes):

- Outcome → Test: Programs that appear to “work” get expanded; programs that appear to fail get cut — or paradoxically expanded
- Outcome → Strategy: Observed outcomes change behavioral responses
- Outcome → Information: What’s learned affects future elicitation
- Outcome → Resource reallocation: Results drive resource decisions
- Outcome → X (Selection): Program outcomes affect who enters the screening pool

3.2 Moderating effects

Moderating effects (pathway interactions):

- Strategy ↔ Classification: Gaming/evasion degrades effective test accuracy.
- Information → Classification: Confessions affect apparent test performance metrics
- Strategy ↔ Information: How targets behave strategically shapes what information the screening process generates, and vice-versa. In polygraph screening, the interrogation process (information pathway) elicits confessions only from applicants who believe the test works and choose to address perceived detection — a strategic calculation, cognitive error, or emotional response. What interrogators then learn from confession patterns feeds back to change interrogation strategy, while applicants who learn what the polygraph interview focuses on or is like can prepare accordingly. In Chat Control, this coupling is largely absent — automated scanning generates no interactive information, and targets’ strategic evasion may produce no usable intelligence feedback. In other contexts, this coupling may manifest as a deskilling cycle: strategic behavioral adaptation (reduced vigilance when AI is present) produces cumulative skill degradation (an information effect), which reinforces the strategic adaptation (increased AI reliance). This cycle is a cross-domain concern. In security screening, it may manifest as analyst over-reliance on algorithmic flags (confirmation bias); in medical screening, as reduced clinical perspicacity when AI is present. The strength of Strategy ↔ Information coupling may be a key structural difference between screening programs that benefit from non-classification pathways and those that do not.

3.3 Erroneous beliefs and ineffective programs

People can erroneously believe programs work and advocate for their expansion:

- As in medical screening more broadly (see §8), cancer screening survivors who were overdiagnosed with non-progressive cancers may wrongly believe their lives were saved; indeed, there are tremendous psychological incentives to believe so. As a consequence, cancer screening programs are hugely popular despite decades of critical advocacy for informed consent about their possible harms.
- Organized advocacy for Chat Control has included funding structures and litigation strategies suggesting organizational rather than grassroots origin (Heat Initiative/Hopewell Fund; Biddle, 2023), complicating interpretation of apparent public support.¹
- Polygraph and other mass screening programs, once entrenched, are notoriously difficult to gut regardless of their scientific evidentiary basis: the “lie detector” mystique persists

¹ The Heat Initiative, which aggressively promoted Chat Control and client-side scanning, is a project of the Hopewell Fund — part of the Arabella Advisors network, which *The Atlantic* described as the “indisputable heavyweight of Democratic dark money” (quoted in Biddle, 2023). Heat Initiative supported a December 2024 class-action lawsuit seeking up to \$1.2 billion in damages from Apple for abandoning its NeuralHash CSAM scanning system (Marsh Law Firm, 2024). Victims participating in such suits likely do not understand themselves as advancing surveillance infrastructure at the possible expense of child safety. This dynamic instantiates the erroneous beliefs mechanism (§3.3): victims sincerely believe they are advancing child safety, while the organizational infrastructure directing their advocacy pursues broader surveillance expansion that may backfire, undermining child safety — a decoupling between individual motivation and structural effect that mass screening programs systematically produce.

despite decades of devastating scientific critique.

This creates positive feedback loops on perceived rather than actual effectiveness—programs expand based on testimonials and political pressure (“but we have to do *something*”), rather than on scientific evidentiary evaluation.

Surveillance-induced guilt without confession

The Information pathway is more complex than simple confession elicitation. Flagged individuals who believe they were caught may act guilty without confessing to authorities, destroying evidence and potentially even themselves. Such guilty behavior may carry some social benefits outside formal criminal justice channels. For innocent flagged individuals, however, such surveillance-induced psychological repercussions represent pure harm with no offsetting benefit of any kind.

3.4 Cross-pathway dynamics

The Classification $\leftrightarrow$ Strategy and Classification $\leftrightarrow$ Information moderating effects are addressed in §3.2; the focus here is the Strategy $\leftrightarrow$ Information coupling, which is the most consequential cross-pathway dynamic for comparative security screening program analysis.

The Strategy and Information pathways are not independent channels from Test to Outcome. Rather, they are coupled: strategic behavior moderates what information the screening process can generate, and information generated through screening feeds back to reshape strategic behavior. This coupling is represented by the Strategy $\leftrightarrow$ Information arrows.

Note that selection effects (e.g., applicants refraining or withdrawing before reaching the interrogation stage of police polygraphs) represent strategy pathway contributions that bypass the information pathway entirely, further complicating decomposition from outcome data alone.

Again, iBorderCtrl occupies an intermediate position: its kiosk interview format has the appearance of an interrogation — potentially activating the bogus pipeline — but the automation strips out the adaptive human elicitation that may make polygraph interrogation effective along the information pathway. It has the form of Strategy $\leftrightarrow$ Information coupling without the substance.

Overall, the strength of this coupling explains why polygraph programs may net benefit through non-classification pathways while Chat Control cannot: the Strategy $\leftrightarrow$ Information coupling that could compensate for classification pathway failure is architecturally absent.

3.5 Limitations

The parameter estimates are transported across contexts (lab-based bogus pipeline effects applied to field screening; street-crime deterrence estimates applied to applicant screening). This transportability problem is a specific instance of the validation problem that structures the proposed cross-domain framework: the bogus pipeline effect was measurable in lab studies because ignorant subjects confessed. But in security screening contexts, such confessions may be less likely due to

different conditions including higher stakes — meaning neither the parameter estimate nor the screening program’s classifications can be validated against an observable answer sheet. The transportability uncertainty is unquantified and probably much larger than the within-study sampling uncertainty reported here.

Gelman and Carpenter (2020) demonstrated that even within a single study and population, treating classification parameters as known rather than uncertain can substantially alter substantive conclusions when conditions are rare. Security applications compound this problem by additionally transporting across contexts which are likely to be more heterogeneous in this domain than in others (lab and field). Functional forms are assumed (largely linear) without adequate theoretical justification. Future work should develop multi-level versions of this paper’s structural equations with within-pathway heterogeneity, apply Bayesian uncertainty propagation (Gelman and Carpenter 2020) to all parameters, conduct formal sensitivity analysis across functional forms, and — where validation data exists (medical applications with observable ground truth) — test whether the single-level model produces valid causal predictions. The current framework is best understood as a qualitative tool for identifying which pathways matter, how they interact, and what the likely signs are along each pathway and in net, rather than a quantitative tool for estimating effect sizes.

3.6 The Aggregation Problem Revisited

The four-pathway framework was motivated by a critique of Fienberg et al.’s single-pathway analysis in the National Academy of Sciences polygraph report: collapsing all possible causal mechanisms into a single classification accuracy question neglects causal structure and produces misleading policy conclusions. At best, it produces only misleadingly overconfident conclusions without causal inferential basis. At worst, it produces completely wrong conclusions.

At the level of paper organization, the strategy pathway is examined before the Bayesian backfire analysis in §5.1, because it represents the strongest possible favorable argument: if deterrence compensates for classification backfire, the net analysis could still support program implementation. Establishing that it cannot across the full parameter space first means the Bayesian results that follow can be read with that compensating argument already foreclosed, so that their practical importance is better known.

At the level of scientific inference, Fienberg’s analysis illustrates what Cartwright (1989) calls a fundamental constraint: no causes in, no causes out. An analysis that encodes only the classification pathway cannot produce conclusions about strategy, information, or resource effects — not because those pathways are absent, but because they were never specified. The four-pathway causal framework is a direct response to this constraint.

While this critique is valid, the proposed causal framework risks replicating the error at a different level of aggregation. Fienberg’s aggregation error was a vertical one of misspecification: ignoring three of four causal pathways, implicitly assuming a *homo economicus* world in which a clean diagnostic test meets rational subjects and neutral institutions — a construction that neglects complex system dynamics including strategic adaptation, belief effects, and resource implications (Simon, 1955; Tversky & Kahneman, 1974; Herley, 2014; see also §3.2–3.5).

The risk here is horizontal: collapsing within-pathway heterogeneity into single aggregate parameters. The strategy pathway’s “net deterrence effect” combines casual offenders who are deterred with sophisticated offenders who evade — two subpopulations with potentially opposite causal contributions. Aggregating across these subpopulations may produce causal parameters that do not correspond to any interventionally manipulable quantity. Rubenstein et al. (2017) formalize this as causal consistency: an aggregate causal model is consistent with a fine-grained model only under specific conditions the analyst must establish rather than assume.

As Rubenstein et al. and others explain, this is the cholesterol problem: LDL and HDL have opposite effects on cardiovascular risk, so aggregating them into “total cholesterol” does not just lose information — it creates a variable that is causally incoherent as a target of intervention. “Net deterrence” in the strategy pathway may be similarly incoherent.

The same problem also applies to the resource reallocation pathway: aggregating national police forensic investigators and specialist prosecutors into a single ‘capacity’ figure obscures that these two groups likely have divergent elasticities. The first, forensic investigative capacity can theoretically be expanded with sufficient resources by expand funding to hire more police investigators. The second, specialist prosecutor pipelines may be more supply-constrained by market and professional factors regardless of available funding (Hartmann², expert interview, May 2025).

In medical screening — where we can often eventually observe ground truth (e.g., in cancer-specific mortality, or histopathology/lab reports) — we can assess whether aggregate parameters produce valid causal predictions. In security screening, it is often more difficult or impossible to obtain such ground truth measures. The validation problem is not merely practical but structural: the condition we are trying to detect (deceptive intent, future misconduct, criminal behavior) is unobservable not just now but in principle, because behavior is endogenous to the screening itself.

Perdomo et al.’s foundational work on performative prediction (2020) formalizes this: once a predictive model is deployed, the population it describes begins adapting to it, shifting the very distribution the model was trained on. Hardt et al. (2016) show this is especially acute under strategic classification — when subjects actively game the system. In security screening, this feedback is not a nuisance — it is constitutive of the system’s causal structure. In medical diagnosis, the prediction “this patient has cancer” does not change whether the patient has cancer. In security screening, the prediction “this person intends to deceive” changes the conditions under which deception will or will not occur — through deterrence, evasion, strategic adaptation, and the psychological effects of being screened. Security screening is not applied statistics on a fixed

² Expert interview data attributed to Hartmann refer to consultations with Markus Hartmann, Leitender Oberstaatsanwalt and Head of the Zentral- und Ansprechstelle Cybercrime Nordrhein-Westfalen (ZAC NRW) at the Generalstaatsanwaltschaft Köln. Structured consultation took place on May 21, 2025, with subsequent clarifying correspondence in March 2026 covering the German investigative workflow (police forensic investigation followed by prosecutorial evaluation, with no separate triage layer), ZAC NRW caseload and staffing, and the limits of cross-jurisdictional population scaling. Mr. Hartmann endorsed the broader research project in a letter of support (ZAC NRW, August 8, 2025). Quoted figures (e.g., the ZAC NRW prosecutor count of 13 handling approximately 14,000 cases annually, the 2024 NRW figure of approximately 2,300 closed investigations following Facebook account hijackings, the 5–7% error rates for AI tools on pre-filtered seized evidence) reflect Mr. Hartmann’s direct professional knowledge of ZAC NRW operations. Where interview data illustrate broader points supported by published sources (e.g., Europol IOCTA 2023 capacity estimates), both are cited.

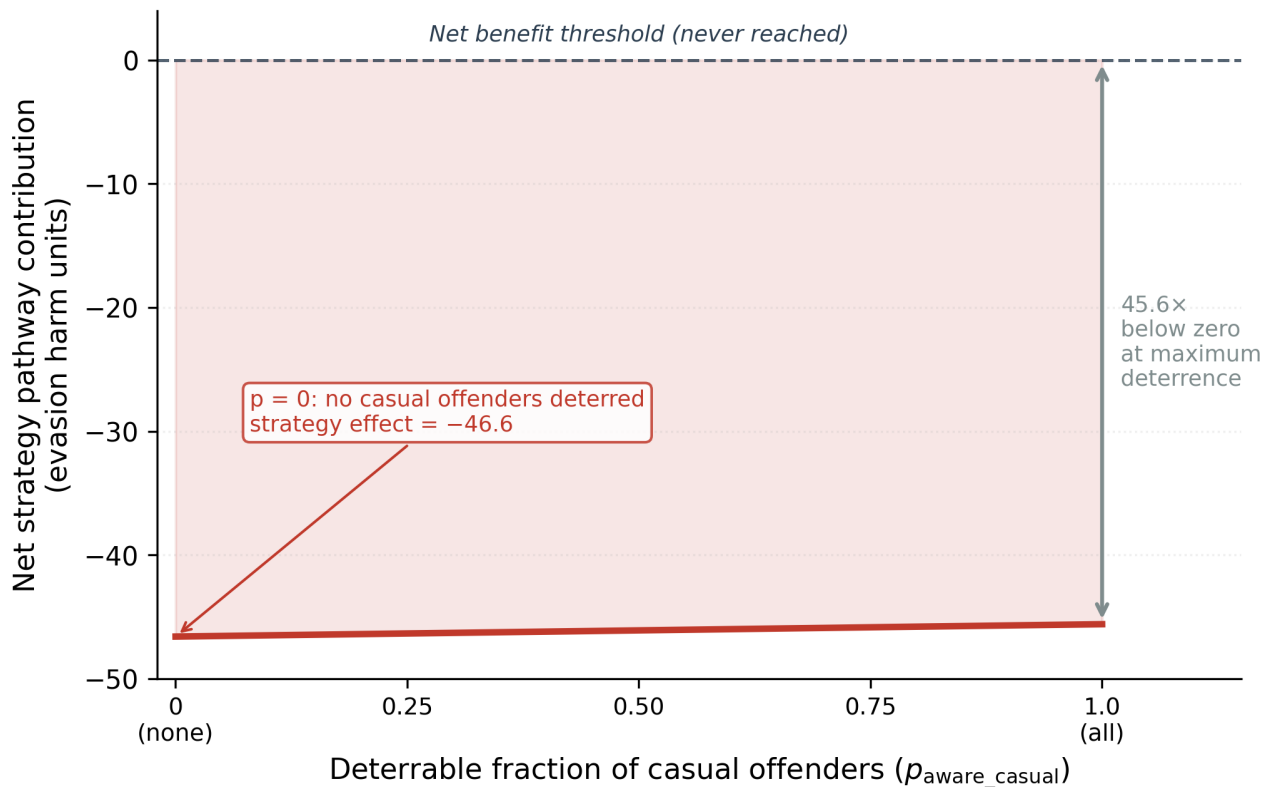


Figure 3. *Chat Control Strategy Pathway: Sensitivity to Casual-Offender Deterrability.* Net strategy pathway contribution as a function of the deterrable fraction of casual offenders ($p_{\text{aware_casual}} \in [0, 1]$). Even assuming all casual offenders are deterrable (right edge), dedicated-offender evasion exceeds casual-offender deterrence by a factor of 46.6 — the line never approaches zero. The shallow slope itself is substantively meaningful: the entire range of uncertainty about casual-offender deterrability moves the outcome by less than 2% of the evasion term. The compensating subpopulation argument is foreclosed across the full parameter space.

population; it is an intervention that reshapes the population it claims to characterize. The minority report problem is not solvable by improving classifier accuracy, because the act of prediction changes the future being predicted.

Any framework — including this one — that requires knowing a priori which subgroups matter and how their causal contributions aggregate faces an epistemological ceiling in security applications that does not apply to medical counterparts. The honest conclusion is not that the four-pathway framework is wrong, but that it is a qualitative diagnostic tool rather than a quantitative predictive one in the security domain, and that cross-domain validation will need to proceed from medical applications toward security applications rather than the reverse. The current paper is an intermediate scaffolding step toward that work.

The methodological response to the horizontal aggregation problem is a decision rule: *disaggregate a pathway when and only when proponents of a program can invoke a compensating subpopulation argument* — i.e., when one subpopulation’s positive pathway contribution might plausibly offset another’s negative contribution in the aggregate. For Chat Control, this argument is live: proponents could claim that casual-offender deterrence compensates for dedicated-offender evasion in the strategy pathway. Disaggregation directly answers that claim.

Both pathways have been disaggregated and implemented: the strategy pathway separates casual offenders into truly unaware and vaguely aware subpopulations alongside dedicated offenders, with sensitivity analysis over the deterrable fraction (disaggregated_strategy_v2.py); the resource pathway separates triage-layer and specialist-layer capacity, showing the specialist bottleneck is a binding constraint regardless of triage scaling (disaggregated_resource_sim.py). That is: the bottleneck presents at the first level in both current practice and proposed possible infrastructure; but even if that resource problem were overcome, another, more intractable one would then present. The overload result appears robust to possible structural changes.

Both files are at github.com/verawilde/equilibrium-screening/security; results are in §5.3–5.5. Under maximally generous assumptions, dedicated-offender evasion still dominates casual-offender deterrence by a factor of 46.6. The net harm result is robust to assumptions about deterrability.

For polygraph programs and iBorderCtrl, no analogous compensating subpopulation argument is central to the proponent case, so disaggregation is not required there.

4. Structural Equations

Following Besserve and Schölkopf 2022's notation, let the system reach equilibrium x^* satisfying self-consistent equations.

4.1 Variables

- Population (X): Number of individuals screened
- Test intensity (T): Screening coverage/thoroughness, range [0,1]
- Classification output (C): Flagged positives, range [0,X]
- Strategic behavior (S): Net deterrence effect (positive = deterred, negative = evasion)
- Information yield (I): Confessions/tips from screening process, range [0,X]
- Resource allocation (R): Proportion of investigative resources on mass vs targeted, range [0,1]
- Outcome (Y): Security/integrity measure (higher = better)
- Base rate (π): True prevalence of target condition, range [0,1]

4.2 Structural Assignments

Classification pathway: $C := f_C(T, X, \pi, \epsilon_C) = T \cdot X \cdot [\pi \cdot \text{sensitivity} + (1-\pi) \cdot (1-\text{specificity})]$

Where sensitivity and specificity are test operating characteristics. Under rarity ($\pi \rightarrow 0$), C is dominated by false positives.

Strategy pathway: $S := f_S(T, C, \theta_S) = \theta_{\text{deterrence}} \cdot T \cdot g_{\text{awareness}}(T) - \theta_{\text{evasion}} \cdot h_{\text{sophistication}}(C)$

Where $g_{\text{awareness}}(T)$ captures how visible/credible the screening program is, $h_{\text{sophistication}}(C)$ captures learning to evade based on classification feedback, and θ parameters capture population characteristics.

Literature on deterrence effects:

- Pratt et al. (2006) meta-analysis: Certainty of apprehension matters more than severity of punishment
- Nagin (2013): “It is the certainty of apprehension, not the severity of the ensuing consequences, that is the more effective deterrent”
- Effect sizes for deterrence are generally modest and uncertain ($r \sim 0.1-0.2$, with wide intervals), and depend heavily on perceived certainty
- Severity of punishment shows effects close to zero with intervals spanning both directions (Doob and Webster 2003)

Information pathway: $I := f_I(T, S, \theta_I) = \theta_{\text{confession}} \cdot T \cdot (1 - S/X) \cdot \text{bogus_pipeline_effect}$

The bogus pipeline effect captures that people confess more when they believe a test works, regardless of whether it actually does.

Literature on bogus pipeline:

- Roese and Jamieson (1993) meta-analysis of 31 studies: $d = .41$, 95% compatibility interval (CI) approximately .25-.57 (positive sign, moderate effect magnitude, meaningful uncertainty)
- Effect is larger when participants must “guess” what the machine detected
- Works by shifting motivation from self-enhancement to self-protection (fear of being caught lying)
- Note: Effect sizes from this literature should be interpreted cautiously given replication concerns in psychology generally

Resource reallocation pathway: $R := f_R(C, I, \theta_R) = \text{base_allocation} + \theta_{\text{false_positive_burden}} \cdot C/\text{capacity} - \theta_{\text{tip_efficiency}} \cdot I$

More false positives drain resources; good tips improve allocation efficiency.

Outcome (equilibrium): $Y^* := f_Y(C, S, I, R, \pi, X) = \theta_C \cdot \text{TP}(C, \pi) - \theta_{\text{FP}} \cdot \text{FP}(C, \pi)$ [classification contribution] + $\theta_S \cdot S$ [strategy contribution] + $\theta_I \cdot I$ [information contribution] - $\theta_R \cdot R$ [opportunity cost] - $\theta_{\text{resource_reallocation_cost}}$

4.3 Feedback Loops

Outcome → Test feedback: $T_{t+1} := f_{\text{feedback}}(T_t, Y_t, \text{perceived_}Y_t, \text{political_pressure})$

Note: perceived_Y may diverge from actual Y due to false positive testimonials, organized advocacy, institutional inertia and sunk costs, and difficulty measuring counterfactual outcomes.

Outcome → Selection feedback: $X_{t+1} := f_{\text{selection}}(X_t, Y_t, \text{program_expansion})$

Strategy ↔ Classification moderation: $\text{effective_sensitivity} := \text{base_sensitivity} \cdot (1 - \text{evasion_factor}(S))$ $\text{effective_specificity} := \text{base_specificity} \cdot (1 - \text{gaming_factor}(S))$

Strategic evasion and gaming degrade test performance.

5. Application 1: Chat Control 2.0

EU Chat Control 2.0 — the Commission’s May 2022 proposal (COM(2022) 209 final) for mandatory AI scanning of all digital communications — is the subject of this analysis. As noted in §1, the current voluntary (CC 1.0) regime uses hash-matching of known CSAM and operates under different statistical logic; the concerns analyzed here apply specifically to the mandatory AI classification proposal.

5.1 Bayesian baseline: frequency-format estimates under favorable assumptions

Following Fienberg et al.’s approach of applying generous assumptions to stress-test a proposed mass screening for a low-prevalence problem’s best-case scenario, we apply Bayes’ Rule to Chat Control using parameters intentionally favorable to proponents: a true positive rate of 90%³, a false positive rate of 16%, and a base rate of 1 CSAM message per 1,000 messages scanned. Scaling to 10 billion messages — a fraction of daily European digital communications — Tables 5-1A and 5-1B show results under two operating thresholds.

Under these unrealistically favorable assumptions, both scenarios reveal the same structural problem from opposite directions: 5-1A maximizes detection but generates 1.6 billion false positives (99.4% of flags are innocent); 5-1B minimizes false positives but leaves 80% of real CSAM undetected. There is no threshold that avoids both failures simultaneously. This is not a calibration problem solvable by adjusting the threshold; it is a mathematical consequence of screening for a rare condition with an imperfect test.

Chat Control result	Non-abusive (innocent)	Abusive (CSAM)	Total
Flagged content	1,598,000,000	9,000,000	1,607,000,000
Not flagged content	8,392,000,000	1,000,000	8,393,000,000
Total	9,990,000,000	10,000,000	10,000,000,000

Table 5.1A. *Hypothetical Estimated Outcomes of Chat Control in High-Sensitivity Mode.* The first of two scenarios with different operating thresholds showing the same structural signal detection problem, illustrating the accuracy-error trade-off when scanning 10 billion messages with a 90% accurate test. This scenario maximizes detection but generates 1.6 billion false positives (99.4% of flags innocent).

³ The 90% true positive rate is generous even relative to optimized targeted tools operating on pre-filtered evidence (see §11); mass screening on unfiltered general communications would be expected to perform substantially worse. Hartmann (expert interview, May 2025) reports 5–7% error rates for AI tools optimized on pre-filtered seized evidence.

Chat Control result	Non-abusive (innocent)	Abusive (CSAM)	Total
Flagged content	39,000,000	2,000,000	41,000,000
Not flagged content	9,951,000,000	8,000,000	9,959,000,000
Total	9,990,000,000	10,000,000	10,000,000,000

Table 5.1B. *Hypothetical Estimated Outcomes of Chat Control in High-Specificity Mode.* The second scenario with the test set to minimize false positives leaves 80% of abuse undetected.

Under the high-sensitivity scenario (Table 5.1A), a flagged message has a 99.4% probability of being innocent — approximately 178 innocent people are flagged for every genuine CSAM message detected. Under the high-specificity scenario (Table 5.1B), the false positive burden falls, but 80% of CSAM evades detection entirely, meaning 8 million abusive messages per 10 billion reach their recipients undetected. There is no threshold at which both failures are avoided simultaneously. This is the accuracy-error trade-off under rarity: not a calibration problem, but a mathematical consequence of the structure of the signal detection problem itself — which is a consequence of the structure of reality and not something better technology can fix.

5.2 Relaxing assumptions: Bayesian uncertainty propagation

The baseline above treats accuracy (90% true positive rate, 84% true negative rate) and base rate (1/1,000) as known. However, they are substantially uncertain. Gelman and Carpenter (2020) demonstrate that treating classification parameters as fixed rather than uncertain can substantially alter substantive conclusions when the target condition is rare. These are the conditions under which Chat Control would operate.

The base rate of CSAM in digital communications is unverified and contested. Mass scanning proposal proponents have strong incentives to inflate it (making the program appear more beneficial) and to inflate novel scanning methods' accuracy (making detection appear more reliable). Independent verification of either parameter is unavailable; the EU Commission's impact assessment (EU Commission 2023) relied on industry-provided figures from Microsoft and Thorn, with no independent verification. In this as in other realms, non-transparency blocks neutral evidentiary evaluation.

Propagating this uncertainty: if the true base rate is 1/10,000 rather than 1/1,000 — plausible given that most CSAM circulates in dedicated encrypted channels rather than general communications — the false-positive-to-true-positive ratio worsens tenfold (from ~178:1 to ~1,778:1), and the true positive rate of the system falls below 0.05%. If accuracy is 80% rather than 90% — still generous — the ratio of false to true positives worsens further. The combination of lower base rate and lower accuracy produces a system that functions almost entirely as a mass surveillance apparatus with negligible child protection benefit.

This sensitivity to parameter uncertainty is not incidental; it is structural. Under conditions of rarity, small changes in base rate assumptions produce large changes in predicted outcomes. Any policy

evaluation that treats these parameters as known rather than uncertain produces overconfidence bias in its program effect estimates. Future work should apply this uncertainty propagation approach to the polygraph and iBorderCtrl parameters; Chat Control serves as the demonstration case because the net harm result is robust even under point estimate assumptions that are intentionally extremely generous to program proponents, making the propagation exercise conservative rather than determinative.

The structural equations in §4 were implemented as a running simulation (available at github.com/verawilde/equilibrium-screening/security); §5.5 presents numeric results propagating uncertainty across all four pathways.

5.3 Four-pathway analysis

Applying the four-pathway framework to Chat Control 2.0, harm dominates along every causal pathway.

Classification: 99.4% of flags are false positives

Strategy: The net deterrence effect is likely near zero or negative for child safety outcomes regardless of deterrence magnitude. Dedicated offenders, who cause disproportionate harm, already use evasion tutorials on specialized Dark Net forums and will go dark.

Among casual offenders, deterrability is heterogeneous: most of those genuinely unaware of illegality cannot be deterred. Their detection is a classification event, not a strategy event. Some vaguely-aware casual offenders may be deterrable through public awareness campaigns accompanying the program. However, harm-weighted disaggregation shows that even under generous assumptions about this deterrable fraction, the dedicated-offender evasion term dominates the aggregate strategy effect by approximately 46.6×. We did not find a parameter combination under which the strategy pathway produces a net positive contribution to Y^* for Chat Control.

Sensitivity analysis over the deterrable fraction of casual offenders ($p_aware_casual \in [0, 1]$) shows that the harm-weighted strategy pathway effect is negative across the full parameter range (see Figure 3, §3.6), with dedicated-offender evasion exceeding maximum casual-offender deterrence by a factor of 46.6.

Information: No interrogation component; no bogus pipeline mechanism; no elicitation $\rightarrow \approx 0$

Resource reallocation: ~1.5B false positives overwhelm ~thousands of investigators $\rightarrow$ Catastrophic

5.4 Resource reallocation: investigative capacity under false positive burden

The classification pathway's prodigious false positive output requires resources to attempt to sort. This creates resource reallocation effects.

Proposition 1 (Resource tractability of known-CSAM scanning).

Hash-matching of known CSAM achieves $Sp \rightarrow 1$ by design. Under these conditions, false positives approach zero and total flags scale with true prevalence: $Flags \approx N \cdot \pi \cdot Se$. The resource

burden is therefore bounded by the actual prevalence of known material in the corpus — a finite quantity, manageable through proportional resource allocation.

Corollary 1 (Resource intractability of unknown-CSAM mass scanning).

For any probabilistic classifier with $Sp < 1$ operating at scale N , the minimum specificity required to remain within budget R satisfies $Sp^* = 1 - R/(c \cdot N \cdot (1 - \pi))$, which approaches 1 as $N \rightarrow \infty$. Since $Sp = 1$ is unachievable for any classifier inferring unknown content from probabilistic signals, no finite investigative budget R satisfies the resource constraint at mass scale. The false positive flood is not an engineering problem solvable by improving classifier accuracy; it is a structural consequence of operating a probabilistic classifier at population scale under rarity. This is what the capacity calculations below instantiate numerically.

Current investigative architecture (two layers). Once a report reaches national law enforcement, there is no lightweight triage stage: police must conduct a full investigation before prosecutorial evaluation. Then, the public prosecutor's office evaluates the case and, where there is sufficient evidence, prepares an indictment. Both layers may already be operating at or near capacity. For example, ZAC NRW, one of Germany's leading cybercrime units, employs 13 prosecutors handling approximately 14,000 cases per year (Hartmann, expert interview, May 2025).

Proposed architecture under Chat Control (three layers). The draft EU Chat Control regulation proposes an additional institutional layer upstream of national law enforcement: a new EU Centre on Child Sexual Abuse. The Commission's description of the EU Centre states it would perform substantive false positive review before forwarding to national authorities — receiving automated reports from service providers, conducting technical verification, and routing confirmed reports to national law enforcement (European Commission, n.d.; Regulation COM (2022) 209 final, Arts. 49-54). This EU Centre would function as a triage and coordination body before cases reach the existing national two-layer pipeline.

Because the three-layer architecture adds an upstream EU Centre as a buffer before national law enforcement, modeling it is the *more generous* test for program proponents. If Chat Control's volume overwhelms even this buffered first layer, then the existing two-layer system — which would absorb the full volume without that buffer — would fare *worse still*. It is therefore logically unnecessary to model the two-layer case separately.

As a rule, this simulation intentionally uses assumptions that are notably generous to Chat Control proponents, following Fienberg's approach of adopting assumptions which were notably favorable to polygraph proponents. The 30-second-per-flag triage assumption applies to this EU Centre layer only, and is an extremely conservative estimate for review of potentially illegal material. Under this generous assumption, EU Centre triage alone consumes millions of person-hours annually (see calculations below).

If cases forwarded from the EU Centre to national police then still require full forensic investigation, as under the current system, then the resource cost per flag is substantially higher than the triage estimate, rendering the overload figures below quite conservative.

Europol IOCTA (2023) estimates approximately 3,000 law enforcement officers across EU member states engaged in CSAM-related work; this serves as the primary capacity estimate. As a corroborating example, ZAC NRW's 13 specialist prosecutors handling ~14,000 cases annually

suggests that even a generous scaling to EU population would not materially change the order of magnitude of the specialist bottleneck (Hartmann, expert interview, May 21, 2025). Institutional heterogeneity across EU member states limits the validity of direct population scaling (Hartmann, expert interview, March 6, 2026). Were the resource overload figures under generous assumptions not so overwhelming, this would warrant further research.

The pre-existing strain on this infrastructure is substantial: NCMEC received approximately 36 million CyberTipline reports in 2023, and the Internet Watch Foundation confirmed approximately 275,000 CSAM URLs the same year — before Chat Control adds any volume. Even under the generous 3,000-figure capacity and assuming only 30 seconds per flag for EU Centre triage, 1.598 billion false positive flags per 10 billion messages represents approximately 13.3 million person-hours — roughly $2.2\times$ total available capacity.

The resource cost of false positives is not hypothetical. In NRW alone, approximately 2,300 investigations against innocent people were opened and subsequently closed in 2024 following a wave of Facebook account hijackings used to post CSAM (Hartmann, expert interview, May 21, 2025). Each required investigative resources — search warrants, cloud data requests, prosecutorial review — before innocence could be established. This is the current baseline false positive burden from existing screening. Chat Control would multiply it by orders of magnitude.

The resource reallocation effect operates in two directions simultaneously: it drains investigative capacity toward mass triage of flagged innocent communications, and away from the targeted investigations of known offenders and networks that currently constitute the most effective child protection work. This is not a trade-off between liberty and security; it is a mechanism by which the program would degrade both.

Framing Chat Control as offering such a trade reflects statistical reification (Greenland, 2017): treating hypothetical data distributions and statistical models as if they reflect known physical laws rather than speculative assumptions for thought experiments, and then mapping those distributions onto values without evidence for the leap. Meanwhile, the confusion matrix outputs — true and false positives and negatives — do not map cleanly onto “security gained” and “liberty lost.” Rather, false positives also carry security costs (e.g., finite investigative resources consumed). True positives, likewise, do not guarantee security benefits proportional to their count. Accepting the liberty-versus-security framing tricks policymakers into arguing about whether to accept a trade that is not on the table, while obscuring that the actual choice may instead be between degrading both values or not implementing the program. The scientific evidence does not establish that we are able to trade some liberty for some security here.

The investigative capacity figure (approximately 3,000 EU CSAM investigators, anchored in Europol IOCTA 2023) is the least precisely estimated parameter in this analysis. However, the net harm result is robust to order-of-magnitude variation: even assuming ten times current capacity, false positive triage would consume the full available caseload, leaving Y^* negative across all simulated parameter combinations. A more precise figure has been requested from practitioner sources; current estimates are conservative anchors from published reports.

5.5 Four-pathway simulation: numeric results

The structural equations in §4 are implemented as a Python simulation (available at github.com/verawilde/equilibrium-screening). Key results are summarized here.

Chat Control ($\pi = 1/1,000$, favorable assumptions): $Y^* = -5.06 \rightarrow$ net harm across all base rate values tested (1/100 to 1/10,000). Resource load $2.2\times$ investigative capacity. Under 10,000 Monte Carlo draws varying sensitivity, specificity, evasion rate, and base rate simultaneously: **P(net harm) = 100%**.

Police Polygraph ($\pi = 1/10$): $Y^* = +0.13 \rightarrow$ modest net benefit under baseline parameters, driven primarily by the information pathway (bogus pipeline, $d = .41$) offsetting negative classification and modest strategy effects.⁴

To assess whether the net harm finding depends on the analyst's normative weighting of pathway contributions, we conducted a sensitivity analysis over theta values, varying θ_C (benefit per true positive: 0.5, 1.0, 2.0), θ_{FP} (harm per false positive: -0.1, -0.5, -1.0), and θ_R (resource overload penalty: -0.5, -2.0, -3.0) across all 27 combinations. θ_S and θ_I were not varied, as Chat Control's near-zero strategy effect and zero information yield mean these weights cannot affect the sign of Y^* .

Result: Y^* is negative in all 27 combinations, ranging from -7.71 to -1.31. Even under the most favorable normative assumptions — high weight on true positives, minimal weight on false positive harm and resource costs — the program produces net harm. The net harm finding appears to be determined along this pathway by the program's structural properties.

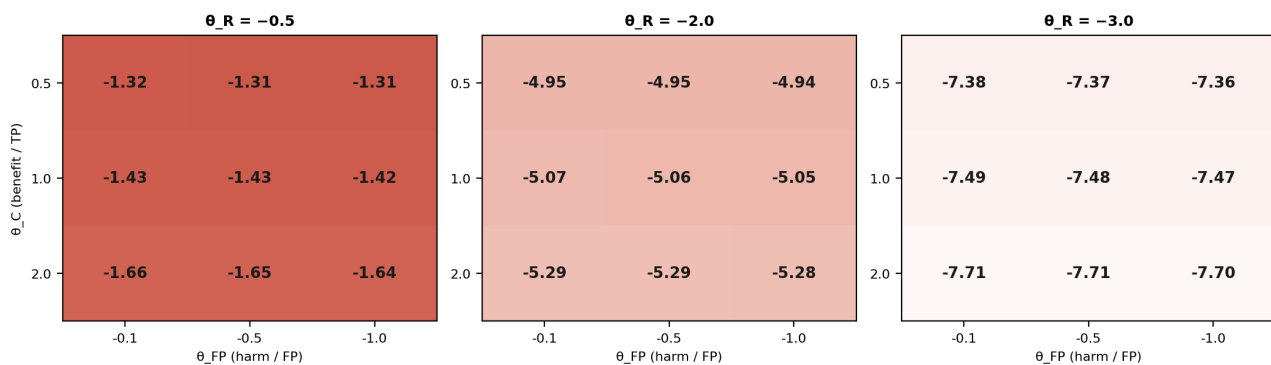


Figure 4. *Normative Sensitivity Analysis: Y^* Under All 27 Theta Combinations.* Each cell shows Y^* under a specific combination of θ_C (benefit per true positive: 0.5, 1.0, 2.0), θ_{FP} (harm per false positive: -0.1, -0.5, -1.0), and θ_R (resource overload penalty: -0.5, -2.0, -3.0). θ_S and θ_I not varied (Chat Control strategy ≈ 0 , information = 0). All 27 combinations yield $Y^* < 0$ (range: -7.71 to -1.31). Net harm appears structurally determined, robust to different normative weightings.

The 3,000-figure spans both layers. The specialist layer is approximately one order of magnitude smaller: scaling ZAC NRW's 13 prosecutors to the EU population suggests roughly 325 specialist prosecutors EU-wide, though such scaling should be treated as a placeholder given institutional heterogeneity across member states. Disaggregating the two layers shows the robustness argument

⁴ The $Y^* = +0.13$ result is robust to base rate uncertainty: sensitivity analysis over π from 1/5 to 1/1,000 produces net benefit throughout, as strategy and information pathway contributions are independent of π in this model. The fragile assumption is not π but the information pathway parameter: the bogus pipeline effect ($d = .41$, Roese & Jamieson 1993) is estimated from laboratory studies under conditions likely more favorable to confession elicitation than operational applicant screening. If the effect does not transport or is substantially attenuated, then Y^* would approach zero or turn negative. Future work should propagate this transportability uncertainty formally, following Gelman and Carpenter (2020).

is conservative, not generous. Even if triage is fully outsourced to non-specialist labor — scaling the first-pass layer indefinitely — the specialist investigation layer remains a binding constraint. At a 1% escalation rate from triage to specialist review (already very low), Chat Control generates ~16 million specialist-level cases per year against a capacity of 325,000: a 49× overload. At 5%, the overload is 246×.

Coping with the workload requires clearing two sequential bottlenecks that cannot be resolved by hiring more triage workers. First, national police forensic investigators must conduct full forensic analysis of every case forwarded from the EU Centre; there is no lightweight substitute for this step under current law (Hartmann, expert interview, March 6, 2026). This presents the earlier and more acute constraint. Second, even if national police forensic capacity were somehow expanded, specialist prosecutors face a structurally different supply constraint: the pipeline of qualified CSAM prosecutors is determined by legal training markets and professional factors, not funding availability. Both bottlenecks must be cleared simultaneously for Chat Control's output to be processed. Neither can be.

5.6 Chat Control equilibrium prediction

The four-pathway analysis converges on a single prediction: Chat Control 2.0 would net harm child safety.

Each pathway contributes to this conclusion independently. Along the classification pathway, applying Bayes' Rule produces estimated hypothetical outcomes in which 99.4% of flags are innocent even under assumptions intentionally favorable to proponents (see §5.1–5.2). Along the strategy pathway, the program fails to deter the offenders who cause disproportionate harm: dedicated criminals already use evasion tools and will migrate to darker infrastructure, while the subset of casual offenders who are genuinely unaware of illegality cannot be deterred by a program they do not know exists or applies to them. Even under maximally generous assumptions about casual-offender deterrability, dedicated-offender evasion dominates the aggregate strategy effect by a factor of 46.6 (§3.6, Figure 3). The information pathway contributes nothing: automated scanning generates no human interaction, no interrogation, no bogus pipeline mechanism. There is no elicitation. And along the resource reallocation pathway, the false positive flood overwhelms investigative capacity at every layer of the proposed architecture. This diverts finite resources away from the targeted investigations of known networks that currently constitute the most effective child protection work.

Unlike police polygraph programs, where strategy and information pathway contributions may compensate for classification pathway failure, Chat Control has no apparent compensating mechanisms. The Strategy↔Information coupling that could offset classification backfire is architecturally absent. All four pathways point in the same direction.

Prediction: Net harm to child safety.

6. Application 2: Police Polygraph Programs

The empirical question is whether polygraph adoption at the department level predicts downstream workforce behavior.⁵ This section operationalizes that as citizen complaints of excessive use of force, a proxy for the kind of officer misconduct polygraph screening intends to select against.

6.1 LEMAS difference-in-differences findings

From U.S. nationally representative Law Enforcement Management and Administrative Statistics (LEMAS) data⁶, 2003-2007, using difference-in-differences with negative binomial regression:

Sustained complaints (use of force):

- IRR: 3.0
- 95% CI .76-11.8
- $p = 0.12$

Total complaints (use of force):

- IRR: 2.4
- 95% CI 1.17-4.97
- $p = 0.02$

Sustained citizen complaints of excessive use of force are rare events: most departments report none in any given survey wave (64% in 2003, 83% in 2007; see Figure 5), making the outcome a zero-inflated count with substantial overdispersion (Pearson $\chi^2/df = 113$). Negative binomial regression is appropriate for this count structure; results are reported as incidence rate ratios (IRR). The treatment group consists of departments that adopted polygraph screening between survey waves ($N \approx 33$ with complete covariate data); the control group consists of departments that never used polygraph screening in either wave. Covariates were log sworn officers and log population served (log-transformed to account for right skew in department size), collective bargaining authorization, internal affairs unit, and Census region.

⁵ The author's prior research on U.S. federal polygraph programs included extensive Freedom of Information Act (FOIA) requests and some subsequent litigation against the CIA, FBI, DoD, and OPM seeking data to assess for possible demographic bias in screening outcomes. *Sack v. Department of Defense* (2012) concerned educational requester status; DOD tried and failed to deny educational requestor status for reduced-fee access available to university researchers for my NSF-sponsored dissertation research. The agencies produced documents but withheld the requested data, precluding the planned observational analysis. The LEMAS analysis reported here uses publicly available data to address a different but related question about program-level outcomes. For the document archive see Wilde, 2016.

⁶ The analysis uses LEMAS 2003 (ICPSR 4411) and 2007 (ICPSR 31161), the only adjacent waves containing both screening tool usage and use-of-force complaint disposition counts. LEMAS 2016 (ICPSR 37323) dropped complaint count variables entirely. LEMAS 2020 (ICPSR 38651) restored them, but the 13-year gap between 2007 and 2020 spans major structural changes in American policing — post-Ferguson reforms, body camera adoption, consent decrees — making the parallel trends assumption indefensible. The hypothesized workforce selection mechanism would also be diluted by roughly two full cycles of officer turnover; we would generally expect a causal effect of an intervention to attenuate substantially over this amount of time.

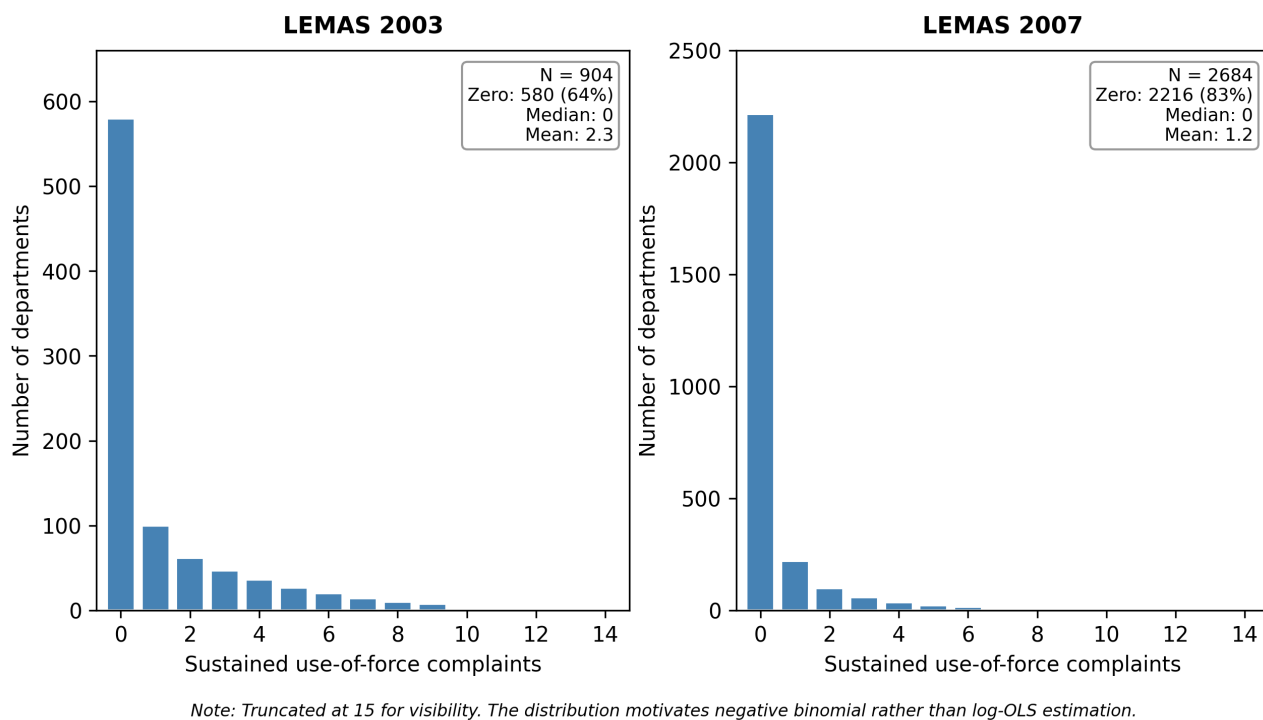


Figure 5. *Distribution of Sustained Use-of-Force Complaints per Department.* The severe zero-inflation, with 64% of departments reporting zero sustained complaints in 2003 and 83% in 2007, motivates negative binomial rather than log-OLS estimation. Truncated at 15 for visibility.

The 95% CI (0.76-11.8) is consistent with as much as a 24% decrease (IRR = 0.76) or an up to 1,080% increase (IRR = 11.8) in sustained complaints, reflecting uncertainty about both direction and magnitude. With a treatment group of $N \approx 33$, the estimate is too unstable to support conclusions about direction. This is consistent with a possible selection-for-non-confessors mechanism operating in reverse: polygraph screening may select officers who are less forthcoming under interrogation, and if sustained complaints require admissions or strong corroboration, effects in either direction are plausible depending on how complaint investigation processes interact with officer characteristics.

Total complaints show IRR = 2.4 (95% CI 1.17–4.97, $p = 0.02$), consistent with as little as a 17% increase (IRR = 1.17) or as much as a 397% increase (IRR = 4.97) in total complaints. Unlike the sustained complaints compatibility interval, which includes both directions, this interval falls entirely above the null: the data are overall consistent with some positive effect on total complaints, though the magnitude is highly uncertain. Point estimates for sustained and total complaints trend in the same direction; but compatibility intervals are wide, N is small, and divergent effects remain compatible with the data. The sustained/total pattern that would most clearly support a non-classification pathway interpretation — fewer sustained complaints alongside unchanged or higher total complaints — can neither be proven nor disproven on the basis of this evidence.

6.2 Interpretation through four-pathway lens

The LEMAS data cannot provide evidence for or against any particular causal pathway dominating police polygraph program net effects. We would need more fine-grained implementation data (e.g.,

independent coders assessing polygraph confessions) for fuller assessment of mechanisms. However, the four-pathway framework structures the space of competing hypotheses:

Hypothesis 1 (Classification/detection): Polygraph testing detects problematic police applicants via physiological measurements, improving workforce quality. The positive IRR (more complaints in adopting departments) does not support this hypothesis. However, the compatibility interval is wide, and we cannot rule out a true negative effect obscured by other causal effects including selection effects. For instance, troubled departments could adopt polygraph programs, which work — but the causes of their troubles that led to program adoption continue or continue trending worse, creating the false appearance of polygraph program failure through common cause.

Hypothesis 2 (Strategy/deterrence): Polygraph screening deters problematic applicants from applying, or prompts self-selection out during the process. At the same time, polygraph screening could also select recruits who are better at not admitting wrongdoing under interrogation; research suggests deceptive subjects with lower socialization scores are more likely to pass polygraphs (Waid, Orne and Wilson 1979). This could produce fewer sustained complaints (which require admissions or strong evidence) without reducing actual misconduct. Overall, the positive IRR similarly does not support this hypothesis, with the same caveat: there could be strategic benefits (deterrence) and harms (self-selection of non-confessing offenders) that are obscured by aggregation errors, small numbers, or other causal effects.

Hypothesis 3 (Information/bogus pipeline): The polygraph questioning process elicits admissions during pre-employment screening that screen out problematic applicants—regardless of chart accuracy. The interview/interrogation, not the physiological measurement, does the work. This is a particularly plausible mechanism for polygraph program efficacy that could be empirically assessed with better data. It suggests a subclass of mass security screenings for low-prevalence problems that may be able to overcome the Bayesian backfire even without access to ground truth to disambiguate true and false positives with minimal secondary screening costs.

Hypothesis 4 (Selection artifact): Departments that adopt polygraph screening may differ from non-adopters in ways not captured by the covariates. For instance, departments under political pressure to address police violence may simultaneously adopt polygraphs and improve complaint investigation processes, producing higher sustained complaint counts through better documentation and investigation rather than worse officer behavior. In this scenario, departments making successful efforts to improve actually wind up looking worse.

Both H1 and H4 are common cause stories, but with different causal ancestors and different implications. In H1, a troubled department is the common cause: it drives both polygraph adoption and worsening misconduct, so complaints rise regardless of whether the program works. In H4, reform pressure is the common cause: it drives both adoption and improved complaint investigation procedures, so documented complaints rise not because misconduct increased but because institutional capacity to record and respond to it improved. The difference matters for interpretation: H1 leaves open the possibility that polygraph adoption has a true benefit masked by confounding; H4 implies the program itself is causally inert but could be misinterpreted as working.

Methods note: Log-transformed OLS is common in the policing literature for complaint outcomes (e.g., Ulkemen 2009; Rydberg and Terrill 2010; Terrill and Ingram 2016). But it is unreliable for zero-inflated counts such as these (64–83% zeros; see Weisburd et al. 2022). Under these conditions, it can produce large coefficients that reflect distributional artifacts rather than treatment

effects (Cameron and Trivedi 2013). Negative binomial regression, which models the count-generating process directly, is more appropriate for overdispersed count data (Cameron and Trivedi 2013; Hougland and Wolf 2016; Shjarback et al. 2021). It models the count-generating process directly and is used here for this reason.

6.3 Four-pathway analysis

Classification: Validity contested; “inconclusives” minimized in practice → Weak/uncertain. LEMAS data do not isolate this or any other causal pathway, leaving effect size, sign, and existence uncertain.

Strategy: Deterrence of problematic applicants and self-selection out of the process are plausible mechanisms, but the LEMAS point estimate (positive IRR) does not support a net deterrence effect. At the same time, the wide compatibility interval and lack of mechanism-level data leave this pathway’s contribution uncertain.

Information: Bogus pipeline effect ($d = .41$; Roese & Jamieson 1993) provides the strongest theoretical basis for a positive program effect: confession elicitation during interrogation screens out problematic applicants regardless of chart accuracy. However, the LEMAS data do not isolate this pathway from the others, and the overall positive IRR does not support the prediction of fewer complaints.

Resource reallocation: Primary screening cost is approximately \$200–500/applicant and does not crowd out patrol resources. However, secondary costs of false positives including additional testing, administrative appeals, legal challenges, and employment costs for wrongly flagged applicants are unquantified and could be substantial given the wide specificity range (.50–.80; NAS 2003). This pathway is unlikely to dominate outcome but cannot be dismissed without better false positive rate estimates at the department level.

Across all four pathways, the data are most consistent with net uncertainty, with the information pathway (bogus pipeline) providing the only strong theoretical basis for a possible net benefit that would seem to require mechanism-level data to confirm or disconfirm.

6.4 Polygraph equilibrium prediction

The simulation (§5.5) predicts $Y^* = +0.13$ for polygraph programs under baseline parameters — a modest net benefit driven primarily by the Information pathway (bogus pipeline). The LEMAS analysis can neither confirm nor disconfirm this prediction: the small treatment group, wide compatibility intervals, and absence of mechanism-specific measures leave the pathway contributions unidentified. The polygraph case illustrates the limits of the framework as an empirical tool in the security domain, where ground truth is often unobservable, while demonstrating its value as a qualitative diagnostic for identifying which pathways matter and how they might interact. It also lays the foundation for future empirical research on decomposing relevant causal effects given those limitations (e.g., using admissions to estimate information effects). Whereas Chat Control converges on harm regardless of such decomposition, polygraph program efficacy cannot be evaluated without it.

7. Application 3: iBorderCtrl (EU Border Security)

iBorderCtrl was an EU Horizon 2020-funded pilot (2016-2019) piloting AI “lie detection” at non-Schengen border crossings in Hungary, Latvia, and Greece (Roussos et al., 2019). The system used facial micro-expression analysis based on Ekman and Friesen’s (1978) action coding system to flag travelers for secondary screening, despite ample evidence that inferring emotional states or deceptive intent from facial movements lacks adequate cross-cultural validation (e.g., Barrett et al., 2019) and is otherwise problematic in signal detection terms (NAS, 2003).

Four-pathway analysis:

Classification: No validated accuracy data; AI emotion detection lacks adequate scientific evidentiary basis → Unknown

Strategy: Sophisticated bad actors (smugglers, traffickers) likely evade; naive travelers flagged → Negative

Information: Some bogus pipeline potential in interviews, but automated kiosk format limits elicitation → Weak

Resource reallocation: Flags require human follow-up; unknown false positive burden → Uncertain

Outcome: The project faced substantial scientific and civil liberties criticism (including from my own work at iborderctrl.no⁷). The technology was not adopted for operational use. This represents a success case for evidence-based critique.

8. The Broader Class: Programs Sharing This Structure

Mass screenings for low-prevalence problems appear across diverse domains:

Security: Population polygraph screening, Chat Control 2.0, iBorderCtrl, terrorist watchlists, airport security

Health: Asymptomatic cancer screening (mammography, PSA, colonoscopy), prenatal testing, newborn screening

Information integrity: Plagiarism detection, AI content detection, misinformation flagging, disinformation detection

Education: Academic integrity monitoring, proctoring AI

Finance: Fraud detection, AML screening

All share an identical mathematical structure as signal detection problems, and face the same accuracy-error tradeoff which leads them to risk incurring invisible net societal damages under

⁷ My work at iBorderCtrl.No has been cited by European Digital Rights (EDRi) as an example of effective science communication in surveillance policy.

conditions of rarity, persistent uncertainty, and secondary screening harms. Yet, they are rarely recognized as a class requiring common analytical treatment.

A further irony: law enforcement agencies themselves rely on encrypted communications for operational security. A Chat Control architecture scanning all digital communications would sweep up their own, a fact noted by practitioners as an important operational concern in addition to the broader liberal democratic one (Hartmann, expert interview, May 21, 2025).

Policy implication: Rather than fighting these programs case-by-case (Whack-A-Mole), society needs to legislate the structure — requiring evidence-based evaluation of programs sharing this dangerous mathematical form before implementation.

Two cases in this paper illustrate that evidence-based critique can succeed: Fienberg's Congressional testimony about his NAS report halted the Department of Energy's proposed mass polygraph screening of all scientists for espionage at the National Labs, and iBorderCtrl was abandoned after much scientific and civil liberties criticism.

9. Comparative Analysis

Scale

- Chat Control: ~450M people, billions of messages
- Police Polygraph: ~thousands of applicants per year
- iBorderCtrl: ~thousands of travelers per day at pilot sites

Base rate

- Chat Control: Extremely low (~0.001%); unverified — industry-provided figures, independent estimates unavailable (see §5.2). Here as across cases, there has been debate about what exactly the problem of interest is (e.g., CSAM transmission versus grooming).
- Police Polygraph: Higher (~5–20%), highly uncertain; no validated prevalence estimate for 'problematic applicants' exists. Range is illustrative and the category again contested (e.g., violent criminals versus other categories of possible offenders).
- iBorderCtrl: Low, unknown — no published base rate estimate for smugglers/traffickers as fraction of border crossers. Again the category could be construed in quite different ways (e.g., people who might intend to cross the border and see if they could stay longer than planned versus terrorists).

Test validity

- Chat Control: AI accuracy inflated/unverifiable
- Police Polygraph: Contested but irrelevant if Information pathway dominates
- iBorderCtrl: No validated accuracy data; AI emotion detection lacks adequate scientific evidentiary basis

Selection effects

- Chat Control: Dedicated offenders migrate to encrypted channels

- Police Polygraph: Problematic applicants self-select out of the process
- iBorderCtrl: Sophisticated actors choose alternative crossings.

Information pathway

- Chat Control: Zero (automated, no human interaction)
- Police Polygraph: Strong (interrogation, bogus pipeline)
- iBorderCtrl: Weak — automated avatar had some adaptive questioning capacity, but the depth of possible automated adaptive elicitation is far shallower than that of human interview/interrogation, potentially limiting bogus pipeline potential

Strategy pathway

- Chat Control: Evasion dominates for sophisticated offenders
- Police Polygraph: Deterrence may dominate for applicant pool
- iBorderCtrl: Sophisticated actors evade; naive travelers flagged

Resource impact

- Chat Control: Catastrophic (billions of false positives)
- Police Polygraph: Manageable (\$200-500/applicant)
- iBorderCtrl: Uncertain (unknown false positive burden at scale)

Feedback dynamics

- Chat Control: Victim advocacy (Heat Initiative) sustains support despite poor evidence
- Police Polygraph: Institutional inertia; “lie detector” mystique in American context
- iBorderCtrl: EU funding cycles sustained pilot despite widespread scientific and social critique

Predicted net effect

- Chat Control: Harm
- Police Polygraph: Uncertain; simulation predicts modest benefit via Information pathway, but LEMAS empirical results inconclusive; mechanism-level data needed
- iBorderCtrl: Likely harm; abandoned after critique

10. Simulation Strategy

The simulation described in §5.5 implements the described framework. Future simulation extensions would build on it using Besserve and Schölkopf’s (2022) deep equilibrium approach, proceeding in three steps:

1. Define cyclic SCM with four pathway structural equations
2. Implement as deep equilibrium layer using Anderson acceleration
3. Parameterize from available data: LEMAS effect sizes for police polygraph (with full uncertainty); bogus pipeline $d = .41$ (.25–.57) from Roese and Jamieson (1993); deterrence effects from criminology meta-analyses (modest, wide intervals); Fienberg-style Bayesian

calculations for classification; sensitivity analysis for unmeasured parameters

A priority empirical extension would be estimating the deterrable fraction of casual offenders ($p_{\text{aware_casual}}$) — the parameter to which the strategy pathway disaggregation is most sensitive in principle, though not in the current analysis. This parameter may in principle be estimable from natural experiments: jurisdictions where public awareness campaigns accompanied high-profile CSAM prosecutions without mandatory scanning could provide before/after variation in casual-offender behavior that might identify the upper bound of the deterrable fraction independently of classification pathway effects. Such estimates might allow the current qualitative foreclosure of the compensating subpopulation argument to be replaced by an empirically anchored one. Access to platform-level data on CSAM report volumes before and after public awareness campaigns, which platforms may currently hold but not share with researchers, could furthermore substantially improve the precision of such an estimate.

Causal interventions to simulate:

- Test intensity intervention (α_T): What happens as screening coverage increases?
- Resource constraint intervention (α_R): How does limited investigative capacity change outcomes?
- Compartmentalized interventions: Can we isolate pathway effects while holding others constant?

Key simulation questions:

1. Under what parameter conditions do programs net benefit vs net harm?
2. How sensitive is the equilibrium to small changes in base rate?
3. Can we identify “tipping points” where feedback loops flip sign?
4. How do Chat Control versus polygraph parameter profiles differ in equilibrium behavior?
5. What happens when perceived effectiveness diverges from actual effectiveness?

11. Parameter Estimates from Literature

Bogus pipeline effect

- Estimate: $d = .41$
- 95% CI .25-.57
- Source: Roese & Jamieson (1993) meta-analysis

Deterrence (certainty — perceived likelihood of being caught)

- Estimate: $r \sim 0.15$
- Plausible range: [0.05, 0.25]
- Source: Pratt et al. (2006); Nagin (2013)

Deterrence (severity — perceived consequences if caught)

- Estimate: $r \sim 0$
- Plausible range: [-0.05, 0.05]

- Source: Doob & Webster (2003); Nagin (2013)

Polygraph sensitivity

- Estimate: 0.70-0.90
- Source: NAS (2003); varies substantially by method

Polygraph specificity

- Estimate: 0.50-0.80
- Source: NAS (2003)

CSAM AI accuracy

- Hash matching (known CSAM only): Industry claims ~99% accuracy, but only detects images already in database — cannot find new CSAM
- AI classifiers (new CSAM): Microsoft claims 88% for grooming detection; Thorn claims >99% for images — no independent verification exists
- Sources: EU Commission (2023) citing Microsoft; Thorn marketing; Steinebach (2023)
- Note on targeted vs. mass screening accuracy: AI systems optimized for precision on seized targeted evidence — data already filtered by reasonable suspicion — show 5-7% error rates in that setting (Hartmann, expert interview, May 21, 2025). Mass screening on unfiltered general communications would be expected to perform substantially worse. The 90% true positive rate used in §5.1 is therefore generous even by the standards of optimized targeted tools.
- Note: Following Fienberg's generous-assumptions approach, our Bayesian analysis uses 90% accuracy. Even under these favorable conditions, the program backfires catastrophically due to base rate rarity.

Polygraph → sustained complaints

- Estimate: IRR = 3.0 (negative binomial)
- 95% CI 0.76-11.8
- Note: Wide interval reflects small treatment group ($N \approx 33$). Direction uncertain.

Polygraph → total complaints

- Estimate: IRR = 2.4 (negative binomial)
- 95% CI 1.17-4.97
- Note: Positive effect.

12. Conclusion

This paper makes four contributions. First, it extends the National Academy of Sciences' single-pathway analysis of mass screening programs (Fienberg et al. 2003) to a four-pathway causal framework encompassing classification, strategy, information, and resource reallocation effects. The key theoretical insight is that programs sharing the mathematical structure of signal detection under rarity, persistent uncertainty, and secondary screening harm operate through multiple causal

channels simultaneously, and that the net effect cannot be determined from classification accuracy alone.

Second, it applies Besserve and Schölkopf's (2022) deep equilibrium framework to formalize the feedback loops through which screening programs reshape the populations they claim to characterize, an instance of Perdomo et al.'s (2020) performative prediction problem. The structural equations and simulation code (github.com/verawilde/equilibrium-screening) provide a reusable toolkit for analyzing other programs in the broader class.

Third, the comparative analysis of Chat Control, police polygraph programs, and iBorderCtrl demonstrates the framework's diagnostic value. For Chat Control, all four pathways converge on net harm: classification backfires under Bayes' Rule, dedicated-offender evasion dominates the strategy pathway even under generous assumptions about casual-offender deterrence, and the information pathway is architecturally absent. iBorderCtrl occupies an intermediate position where weak coupling appears likely insufficient to net compensate for the Bayesian backfire and resource reallocation costs.

This convergence is robust to normative weighting of pathway contributions (§5.5) and to disaggregation of within-pathway heterogeneity (§3.6). The disaggregation decision rule was formulated because proponents could invoke a compensating subpopulation argument for Chat Control: the disaggregated simulation closes that argument empirically rather than by assumption.

For police polygraph programs, the pathways do not provably converge on a net effect of one sign or another. Simulation predicts modest net benefit via the information pathway, but LEMAS empirical analysis with appropriate count models produces inconclusive results with wide compatibility intervals. For iBorderCtrl, the absence of validated accuracy data and the weak information pathway suggest net harm, consistent with the program's abandonment after scientific critique.

Fourth, the paper identifies a structural regularity across apparently unrelated policy domains — security screening, medical screening, information integrity, education, and finance — and argues that the recurring policy failure of implementing programs that backfire under conditions of rarity, uncertainty, and secondary screening harm necessitates structural legislation rather than case-by-case advocacy.

The framework has important limitations (§3.5–3.6). It is better understood as a qualitative diagnostic tool than a quantitative predictive one, particularly in the security domain where ground truth is often persistently unobservable. The aggregation problem of collapsing within-pathway heterogeneity into single parameters is partially addressed by the suggested disaggregation decision rule and implemented simulations, but remains a persistent challenge. Future work should pursue cross-domain validation proceeding from medical applications, where ground truth is relatively potentially observable and many high-quality datasets accessible, toward security applications where they are not.

Additional methodological extensions include searching for multiple equilibria using deflated Newton methods (Farrell et al. 2015), which could reveal whether screening programs have alternative stable states that current parameter estimates might miss.

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Data availability statement: All code and simulation data supporting this analysis are publicly available at github.com/verawilde/equilibrium-screening. The LEMAS datasets are publicly available from ICPSR (study numbers 4411 for 2003 and 31161 for 2007). No new human-subjects data were collected for this analysis.

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